Neuroscientific Frontiers — Module 07: Communication — Study Questions

1. Name the five key regions of the social brain network and specify what aspect of social inference each implements.
2. How does medial prefrontal cortex (mPFC) generate predictions about others' mental states? What is the difference between dorsal and ventral mPFC in explicit vs. implicit mentalizing?
3. How does the right TPJ contribute to self-other distinction? Why is damage to rTPJ devastating for theory of mind?
4. How does Active Inference reinterpret mirror neurons? What is the fundamental difference between "motor resonance" and "shared generative model" accounts?
5. What evidence argues against the classical mirror neuron theory and in favor of the predictive reinterpretation? (List at least three findings.)
6. What is interbrain synchronization? What does hyperscanning reveal about neural coupling during conversation?
7. How does Active Inference explain interpersonal neural coupling? What does "mutual prediction" mean at the neural level?
8. What are leader-follower dynamics in neural synchronization? How does the "lead brain" relate to who is driving the communicative exchange?
9. Trace the hierarchical predictive processing of language from phonemes (STG) to syntax (Broca's area) to discourse (prefrontal/temporal poles). What ERP component indexes prediction error at each level?
10. How does the N400 ERP component demonstrate word-level prediction error? What factors modulate N400 amplitude?
11. How does the P600 demonstrate syntactic prediction error? Compare the P600 with the N400 — what key differences reveal their distinct computational roles?
12. How does oxytocin function as a precision modulator for social prediction errors? Why does this explain both prosocial and potentially hostile behaviors?
13. How do in-group/out-group effects of oxytocin relate to the structure of the social generative model? Does oxytocin make you "nicer" or "more in-group biased"?

14. How does autism relate to social prediction precision? What specific social prediction challenges does inflexibly high precision produce?
15. Compare the predictive model of language comprehension with classical models (Broca-Wernicke). What does predictive processing add that the classical model lacks?
16. How does infant language acquisition reflect the development of a linguistic generative model? What changes in the child's predictive architecture enable vocabulary explosion?
17. What is the neural basis of pragmatic inference (understanding irony, sarcasm, indirect speech acts)? How does the brain predict the speaker's intended meaning beyond literal content?
18. How might Active Inference explain the uncanny valley effect? What happens to social prediction when a stimulus is almost-but-not-quite human?
19. How does the social brain network interact with the self-model network (Module 02)? Where do self-modeling and other-modeling share neural resources, and where do they diverge?
20. Critically evaluate: Does interbrain synchronization reflect genuine information transfer between brains, or is it simply the result of both brains responding to the same stimulus? What experiment could distinguish these?